

Homework 3: Kernel and Non-Kernel Learning Under Real-World Data Imperfections

Course: Graduate Machine Learning

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Assignment Overview

This assignment represents a paradigm shift from traditional algorithmic implementation. While you will build models from scratch, the primary challenge is not the code itself, but rather understanding **when the dataset matters more than the algorithm**.

Real-world data is inherently flawed. You must investigate how different data characteristics, imperfections, and preprocessing decisions dictate the success or failure of Linear Models, Kernel Methods, and Distance-Based Methods.

Grading Emphasis:

- **60%** Data Engineering, Investigation, and Deep Analysis
- **40%** Algorithmic Implementation and Modeling

Section 1: The Four Real-World Manifolds

You must process, engineer, and evaluate models on all four of the following datasets. Each represents a fundamentally different topological manifold and operational challenge. You cannot apply a universal pipeline; each requires a bespoke data engineering strategy.

Dataset 1: Airbnb Open Data

- **Source:** Inside Airbnb Dataset
- **Prediction Targets:** Price (Regression) and Superhost Status (Classification)
- **Engineering Requirements:** You must justify your missing value imputation strategies, explain your criteria for dropping features, and defend your outlier handling methodology.

Dataset 2: NYC 311 Service Requests

- **Source:** NYC Open Data
- **Prediction Targets:** Complaint Category (Classification) and Resolution Time (Regression)
- **Engineering Requirements:** You must sample the data intelligently to maintain distribution integrity, resolve duplicate reports, and engineer meaningful temporal features.

Dataset 3: IBM HR Employee Attrition

- **Source:** IBM HR Analytics Dataset (UCI/Kaggle)
- **Prediction Target:** Employee Attrition (Classification)
- **Engineering Requirements:** You must investigate the root drivers of attrition, apply rigorous feature selection, and implement sampling strategies (e.g., SMOTE, undersampling) to combat imbalance.

Dataset 4: Online Retail Transactions

- **Source:** UCI Machine Learning Repository
- **Prediction Targets:** Customer Segment (Classification) and Future Spending (Regression)
- **Engineering Requirements:** You must transition from raw transactional logs to aggregated, customer-level behavioral features before modeling.

Section 2: Data Quality Investigation (25% of Grade)

Before passing any data to your models, you must conduct and document a rigorous data quality investigation for **each dataset**.

- **Missing Data Analysis:** Categorize and prove the nature of the missingness (MCAR, MAR, or MNAR) and defend your imputation math.
- **Outlier Analysis:** Compare and contrast the effectiveness of Z-score, IQR, and Isolation Forest anomaly detection on your specific distributions.
- **Feature Quality Analysis:** Identify and mitigate redundant features, data leakage, and near-constant features.
- **Label Quality Analysis:** Document potential annotation errors, ambiguous ground truths, and historical data collection bias.

Section 3: Algorithmic Implementation (15% of Grade)

You are prohibited from using scikit-learn or similar libraries for the core model implementations. You may only use NumPy and SciPy. You must implement the following from scratch for all four datasets.

Non-Kernel Methods:

- Logistic Regression
- Linear Regression
- K-Nearest Neighbors (KNN)
- Decision Tree

Kernel Methods:

- Kernel Support Vector Machine (SVM)
- Kernel Ridge Regression (KRR)
- Kernel KNN
- Kernel Principal Component Analysis (KPCA) + downstream classifier

Section 4: Kernel Investigation (30% of Grade)

Reporting raw metrics is insufficient. You must provide a mathematical and dataset-specific justification for **why** one kernel or algorithm outperformed another.

- **Prohibited Phrasing:** "The RBF kernel got 92% accuracy, which is the best."
- **Expected Phrasing:** "The RBF kernel succeeded because the transformed feature space better captured the nonlinear, localized interactions created by the geographic coordinates and temporal variables in Dataset 2."

Section 5: Research-Level Discussion (30% of Grade)

This section tests your ability to synthesize findings across multiple domains. Your final report must address the following discussion points thoroughly.

Discussion A: Cross-Dataset Comparison

Compare the four manifolds directly. Identify which dataset was the hardest to clean, which required the most aggressive feature engineering, which benefited the most from the "kernel trick," and which benefited the least.

Discussion B: Failure Analysis

For every dataset, isolate your model's **10 worst predictions**. Conduct a post-mortem to explain exactly why the model failed on these instances, and determine whether the failure was a systemic model limitation or an artifact of the data quality.

Discussion C: Bias and Fairness

Investigate the underlying distributions for sampling bias, temporal drift, and geographic bias. Discuss how these biases alter the model's real-world reliability.

Discussion D: Computational Analysis

Compare the training time and memory usage of your models across the datasets. Discuss the hardware limitations of the $O(n^2)$ and $O(n^3)$ complexities inherent in generating and inverting kernel matrices.

Discussion E: Theory vs. Reality

Using direct evidence from your experiments on these four datasets, answer the following:
Why do textbook datasets (like Iris or standard MNIST) consistently overestimate model performance and fail to prepare engineers for production machine learning?

Section 6: Final Deliverable Format

Your submission must be structured as a **Mini-Conference Paper**. Do not submit a raw, unformatted Jupyter Notebook as your primary document.

Format Requirements:

- **Length:** 20 to 35 pages (excluding code appendix).
- **Structure:**
 1. Introduction
 2. Dataset Description & Topology
 3. Data Cleaning & Feature Engineering
 4. Experimental Setup & Model Development
 5. Results
 6. Error & Failure Analysis
 7. Cross-Dataset Comparison
 8. Discussion (Bias, Computation, Theory vs. Reality)
 9. Conclusion
- **Code Submission:** Attach your fully commented, object-oriented Python scripts or Jupyter Notebooks as a separate zipped appendix.

DATASET LINKS

1. Airbnb Open Data (Inside Airbnb)
<https://insideairbnb.com/get-the-data/>
2. NYC 311 Service Requests
<https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9>
3. IBM HR Employee Attrition
<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>
4. Online Retail Transactions (UCI)
<https://archive.ics.uci.edu/dataset/352/online+retail>

Direct Excel File:

<https://archive.ics.uci.edu/ml/machine-learning-databases/00352/Online%20Retail.xlsx>